

KARUPPASAMY M

Software Developer | Full-Stack & AI Application Development
Trichy, Tamil Nadu, India | +91 82703 74293 | karuppasamy.murugesan02@gmail.com
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Diploma Computer Science Engineering student and hands-on software developer who builds and deploys full-stack web applications, AI-powered tools and business software. Works across React, JavaScript/TypeScript and Python, with practical experience in LLM applications, data analysis, and Linux and networking fundamentals. Delivered client software including an ERP system, an e-commerce site and billing software, alongside AI coding, legal research and resume-automation products. Active in technical webinars, product development and student mentoring.

TECHNICAL SKILLS

Programming: Python, JavaScript (ES6+), TypeScript, C, C++, SQL

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3, React Router, Zustand, Responsive Web Design

Backend & Database: Node.js, Express.js, REST APIs, Supabase (PostgreSQL), MySQL, Firebase, MongoDB

AI & Machine Learning: OpenAI API, Google Gemini, Groq, OpenRouter, AI agents, LLM applications, prompt engineering, TensorFlow/Keras, pandas, NumPy, scikit-learn, Streamlit

Systems & Security: Linux, Unix, Kali Linux, networking fundamentals (TCP/IP, DNS, HTTP/HTTPS), cybersecurity and penetration testing fundamentals

Tools & DevOps: Git, GitHub, Vercel, Render, Postman, npm, Docker

Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS

EXPERIENCE

Google Student Ambassador — Google Student Ambassadors (India)

Tiruchirappalli, Tamil Nadu | May 2026 – Present

- Selected as a campus ambassador to represent Google developer tools and AI technologies within the student developer community.
- Support student-facing technical sessions and awareness activities on AI and Google developer platforms.

AI & Machine Learning Intern — VDart Academy

Tiruchirappalli, Tamil Nadu | Apr 2026 – May 2026

- Completed a structured industry internship in artificial intelligence and machine learning application development.

Data Analyst, Machine Learning & AI Intern — NoviTech R&D Pvt Ltd

Coimbatore, Tamil Nadu | Sep 2025 – Jan 2026

- Completed a five-month multi-domain internship across three consecutive tracks: Data Analyst (Sep–Oct 2025), Machine Learning (Oct–Nov 2025) and AI (Dec 2025–Jan 2026).
- Progressed from exploratory data analysis and reporting on business datasets, to building and evaluating machine learning models, to applied AI implementation work.

EDUCATION

Diploma in Computer Science Engineering | GPA: 9.0 / 10 (90%)

Seshasayee Institute of Technology, Trichy, Tamil Nadu | Jun 2025 – Jun 2027 (Expected)

Higher Secondary Certificate (Class 12), Computer Science | 74%

Government Higher Secondary School, Guziliamparai, Dindigul, Tamil Nadu | Completed Apr 2025

SELECTED PROJECTS

Agent K (Speed Agent) — Browser-Based AI Coding Agent & IDE | *React, Vite, Zustand, Monaco Editor, Supabase, Tailwind CSS* | [GitHub](#) | [Live Demo](#)

- Built an AI coding agent and browser-based IDE concept, inspired by modern AI developer environments, that creates and modifies project files from natural-language chat instructions.
- Integrated a multi-model switcher across OpenAI, Google Gemini, Groq and OpenRouter, and implemented a multi-tab Monaco editor with Zustand state management and an agent response parser that applies generated code to the project tree.
- Designed the Supabase backend — authentication, project storage and Row-Level Security policies — keeping each user's projects isolated in a multi-tenant architecture.

Pickup Law Agent — AI Legal Research & Learning Assistant | *React, TypeScript, Vite, Tailwind CSS, LLM API, SQL* | [GitHub](#) | [Live Demo](#)

- Developed and deployed an AI-assisted research tool that helps law students look up and understand Indian statutory provisions; scoped as an educational research aid, not legal advice.
- Structured a legal knowledge base of Indian Penal Code sections, Code of Criminal Procedure and Constitution question-answer datasets in CSV/JSON to ground the assistant's responses.
- Implemented role-based agent behaviour so responses adapt to user context, layering prompt templates over an LLM API with a SQL schema for persisted data.

AI Resume Builder | *React, Tailwind CSS, OpenAI API, Vercel* | [Live Demo](#)

- Engineered an AI-powered resume builder using the OpenAI API with custom prompt engineering to generate ATS-optimised, role-specific resume content, deployed and in production use.
- Designed industry-specific prompt templates and a real-time two-panel live preview giving users instant formatting feedback, with browser-based PDF export producing clean, ATS-parseable documents.

CLIENT PROJECTS

AC Duct ERP Management System | *React, Tailwind CSS, Firebase, MySQL* | *Private repository*

- Delivered a custom ERP system solo for an AC duct fabrication business over 3–4 months, owning requirement gathering, UI design, implementation, testing and handover.
- Built work-order and project tracking, client/vendor records with payment status, and a business insights dashboard, replacing the client's manual spreadsheet workflow.
- Shipped a mobile-responsive interface so field staff could update job progress on site without a desktop machine.

RV Prethivk Furniture Shop — E-Commerce Website | *Client Project*

- Designed and delivered a customer-facing e-commerce website for a furniture retail business, presenting the shop's product catalogue online for customer browsing.

RV Prethivk Furniture Shop — Billing Software | *Client Project*

- Developed billing software for the same client, digitising invoice generation and day-to-day sales transaction records for counter staff.

ADDITIONAL PROJECTS

DeepVOG Light — Edge Gaze-Estimation CNN | *Python, TensorFlow/Keras, TensorFlow Lite, ARM Ethos-U (Vela), C++* | [GitHub](#)

- Built a lightweight convolutional neural network in TensorFlow/Keras that predicts gaze coordinates from 96×96 grayscale eye images, sized to run on microcontroller-class hardware.
- Applied full INT8 quantisation and compiled the model with the ARM Vela toolchain for the Ethos-U55 NPU — a 114 KB model within a 176 KB peak-SRAM budget — and automated the train-quantise-export pipeline into a single build script that emits an embedded C array for firmware integration.

Butter MV — Streaming-Style Media Web Application | *React, Vite, JavaScript, Tailwind CSS, Vercel* | [GitHub](#) | [Live Demo](#)

- Built a streaming-platform-style front end exploring how catalogue browsing and playback can be delivered on a lightweight, low-cost architecture instead of heavy media infrastructure.
- Implemented the catalogue UI and playback interface in React with Vite, applying code splitting and lazy loading, and explored segmented media delivery to cut storage and bandwidth cost per stream.

AI Sales Intelligence | *Python, Streamlit, pandas, NumPy, scikit-learn, Plotly* | *Not deployed* | [GitHub](#)

- Built a sales intelligence prototype that ingests business datasets and surfaces trends and performance breakdowns, using pandas and scikit-learn for analysis and Plotly visualisations in an interactive Streamlit dashboard.

AI-Powered Local E-Commerce Platform | *React, Tailwind CSS, Supabase* | [Live Demo](#) — Multi-vendor platform with product listings, cart, checkout, vendor dashboard and a rule-based recommendation engine.

HRMS — HR Analytics & Reporting | *Python, Streamlit, pandas, scikit-learn, Plotly* | *Not deployed* | [GitHub](#) — Processes employee datasets into workforce analytics dashboards and automated PDF reports.

Webinar Management System | *React, Tailwind CSS, Supabase* | [Live Demo](#) — Event scheduling, attendee registration with capacity control, admin dashboard and client-side reminders.

TECHNICAL LEARNING

60-Day Data Structures & Algorithms Program | *Python, C++* | [GitHub](#) — Daily problem-solving across core data structures, searching, sorting and recursion with complexity analysis.

60-Day React.js Implementation Program | *React, JavaScript, Vite* | [GitHub](#) — Sequence of React applications covering hooks, routing, API integration and reusable UI patterns.

TECHNICAL COMMUNITY & PRODUCT ACTIVITIES

- Participate in and support online technical webinars covering software development, artificial intelligence, startups and emerging technologies, and mentor students through technical guidance and project discussions.
- Work on product ideation, development and demonstration, including technical marketing and presentation of AI and web products to non-technical audiences.

CERTIFICATIONS

- Google Core Python — Nov 2025 | Google 5-Day Generative AI Agents Intensive — Dec 2025
- Machine Learning Fundamentals — Oct 2025 | AI Basics, Infosys Springboard — Oct 2025
- Full-Stack AI Engineer 2026 (Machine Learning, Deep Learning, Generative AI) | Cloud & DevOps

ADDITIONAL INFORMATION

Soft Skills: Technical communication, presentation, product demonstration, student mentoring, collaboration, problem solving

Languages: English — professional working proficiency | Tamil — native **GitHub:** 45+ public repositories across web, AI and Python.